Assignment No.2 (Course STA 100)

Spring 2024

By M.Kashif(03064963933)

Question 1: Marks: 03+02+02+03

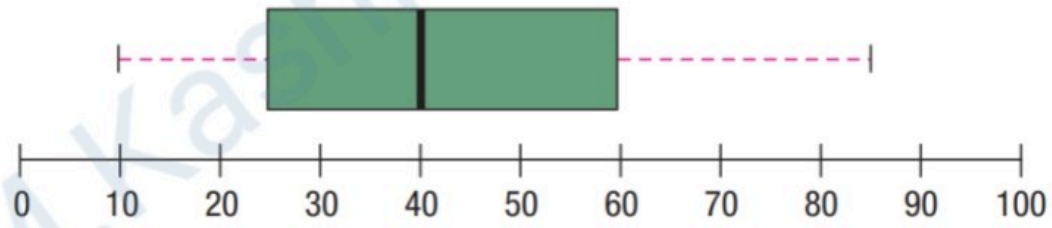
a) From a student club consisting of 50 people. How many different choices of officers are possible if:

i. Three members are chosen randomly.
ii. A president and a treasurer are to be chosen.

b) In one year, three awards (research, teaching, and service) will be given to a class of 25 graduate students in a statistics department. If each student can receive at most one award, how many possible selections are there?

c) A young boy asks his mother to get 5 Game-Boy™ cartridges from his collection of 10 arcade and 5 sports games. How many ways are there that his mother can get 3 arcades and 2 sports games?

d) The box plot below shows the amount spent for books and supplies per year by students at four-year public colleges.



i. Estimate the median amount spent.
ii. Estimate the first and third quartiles for the amount spent.
iii. Estimate the interquartile range for the amount spent.
iv. Beyond what point is a value considered an outlier?
v. Identify any outliers and estimate their value.
vi. Is the distribution symmetrical or positively or negatively skewed?

Solution:

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Part a

From a student club consisting of 50 people:

i. Three members are chosen randomly

The number of ways to choose 3 members out of 50 can be calculated using the combination formula:

$$C = \frac{n!}{r!(n-r)!}$$

$$n = 50$$

$$r = 3$$

$${}^{50}C_3 = \frac{50!}{3!(50-3)!}$$

$${}^{50}C_3 = \frac{50.49.48.47!}{3.2.1!.47!}$$

$${}^{50}C_3 = \frac{117600}{6}$$

$${}^{50}C_3 = 19600$$

So, the number of different choices of 3 members is 19,600.

ii. A president and a treasurer are to be chosen

permutation is used:

$$P = \frac{n!}{(n-r)!}$$

$$n = 50$$

$$r = 2$$

$${}^{50}P_2 = \frac{50!}{(50-2)!}$$

$${}^{50}P_2 = \frac{50.49.48!}{48!}$$

$${}^{50}P_2 = 2450$$

So, the number of different choices to choose a president and a treasurer is 2,450.

Part b

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In one year, three awards (research, teaching, and service) will be given to a class of 25 graduate students, with each student receiving at most one award.

$$P = \frac{n!}{(n-r)!}$$
$$n = 25$$
$$r = 2$$
$${}^{25}P_3 = \frac{25!}{(25-3)!}$$
$${}^{25}P_3 = \frac{25.24.23.22!}{22!}$$
$${}^{25}P_3 = 13800$$

So, the number of possible selections is 13,800.

Part c

A young boy asks his mother to get 5 Game-Boy™ cartridges from his collection of 10 arcade and 5 sports games. The number of ways to get 3 arcade games out of 10 and 2 sports games out of 5 is calculated as:

$$Y = 23.39 + X = 0.23$$

$$C = \frac{n!}{r!(n-r)!}$$
$$n = 5$$
$$r = 2$$
$${}^{5}C_2 = \frac{5!}{2!(5-2)!}$$
$${}^{10}C_3 = \frac{5.4.3!}{2.1!3!}$$
$${}^{10}C_3 = \frac{20}{2}$$
$${}^{10}C_3 = 10$$

The total number of ways is:

$=120\times10$

$=1200$

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So, the number of ways his mother can get 3 arcade and 2 sports games is 1,200.

Part d

Box Plot Analysis

i. Estimate the median amount spent

The median is the middle value in the box plot, so estimate this value from the given plot is 40.

ii. Estimate the first and third quartiles for the amount spent

Estimate Q1=25 and Q3=65 from the left and right edges of the box in the plot.

iii.Estimate the interquartile range for the amount spent

$$IQR = Q_3 - Q_1$$
$$IQR = 65 - 25$$
$$IQR = 40$$

iv. Beyond what point is a value considered an outlier?

$$IQR = 40$$
$$Q_1 = 25$$
$$Q_3 = 65$$
$$Q_1 - 1.5 * IQR$$
$$= 25 - 1.5 * 40$$
$$= -35$$
$$Q_3 + 1.5 * IQR$$
$$= 65 + 1.5 * 40$$
$$= 125$$

v. Identify any outliers and estimate their value

Based on the box plot:

There are no data points outside the whiskers, which range from approximately 10 to 80.

Therefore, there are no outliers, as all data points fall within the range −35 to 125.

vi. Is the distribution symmetrical or positively or negatively skewed?

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The distribution appears to be approximately symmetrical, as the median line is roughly in the center of the box and the whiskers on both sides are of similar length.

Question 2: Marks: 01+03+03+01+02

The sales manager of North American Copier Sales, which has a large sales force throughout the United States and Canada, wants to determine whether there is a relationship between the number of sales calls made in a month and the number of copiers sold that month. The manager selects a random sample of 15 representatives and determines, for each representative, the number of sales calls made, and the number of copiers sold. This information is reported in the Table given below.

Sale Representative	Sales Calls (x)	Copiers Sold (y)
Brian Virost	96	41
Carlos Ramirez	40	41
Carol Saia	104	51
Greg Fish	128	60
Jeff Hall	164	61
Mark Reynolds	76	29
Meryl Rumsey	72	39
Mike Kiel	80	50
Ray Snarsky	36	28
Rich Niles	84	43
Ron Broderick	180	70
Sal Spina	132	56
Soni Jones	120	45
Susan Welch	44	31
Tom Keller	84	30

a) Develop a scatter diagram to display the relationship between sales calls and copier sold.
b) Determine the simple linear regression equation ($Y = a + bx$).

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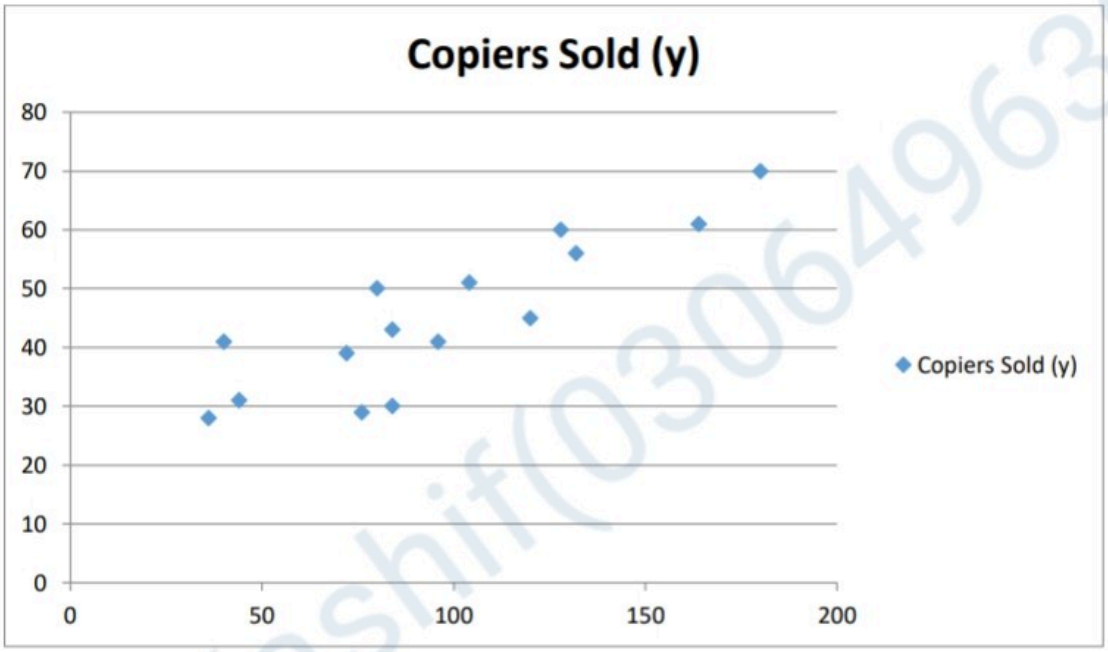
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c) Determine the correlation coefficient.
d) Interpret the strength of the correlation coefficient.
e) Calculate and interpret the standard error of estimate.

Solution:

Scatter Diagram

The scatter diagram displays the relationship between the number of sales calls made and the number of copiers sold.



Simple Linear Regression Equation(Y=a+bX)

To determine the linear regression equation, we find the slop of b and a;

$n = 15$

$\sum x = 1440$

$\sum y = 675$

$\sum x^2 = 154608$

$\sum y^2 = 31625$

$\sum xy = 72364$

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Calculating the slop b:

Formula

$$b=\frac{n\left(\sum xy\right)-\left(\sum x\right)\left(\sum y\right)}{n\left(\sum x^2\right)-\left(\sum x\right)^2}$$

$$b=\frac{15(72364)-(1440)(675)}{15(154608)-(1440)^2}$$

$$b=\frac{1085460-972000}{2319120-2073600}$$

$$b=\frac{113460}{245520}$$

$$b=0.462$$

Calculating the slop a:

Formula:

$$a=\frac{\left(\sum y\right)\left(\sum x^2\right)-\left(\sum x\right)\left(\sum xy\right)}{n\left(\sum x^2\right)-\left(\sum x\right)^2}$$

$$a=\frac{(675)(154608)-(1440)(72364)}{15(154608)-(1440)^2}$$

$$a=\frac{104347200-104182560}{245520}$$

$$a=\frac{164640}{245520}$$

$$a\approx 0.67$$

The Regression Equation is:

$$Y=0.66+0.462X$$

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Correlation Coefficient r

The correlation coefficient r is calculated by:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{\left[n(\sum x^2) - (\sum x)^2\right]\left[n(\sum y^2) - (\sum y)^2\right]}}$$

Calculation:

$$r = \frac{15(72364) - (1440)(675)}{\sqrt{\left[15(154608) - (1440)^2\right]\left[n(31625) - (675)^2\right]}}$$

$$r = \frac{1085460 - 972000}{\sqrt{245520.18750}}$$

$$r = \frac{113460}{\sqrt{4603500000}}$$

$$r = \frac{113460}{67840.78}$$

$$r = 1.67$$

The correlation coefficient obtained by earlier regression function r=0.8646.

Interpretation of the Correlation Coefficient

The correlation coefficient of 0.8646 suggests a strong positive relationship between the number of sales calls and the number of copiers sold.

Standard Error of Estimate

The standard error of estimate (SE) measure the accuracy of predictions made with regression equation.

It is calculated as follow:

$$SE = \frac{\sum\left(Y_i - \hat{Y}_i\right)^2}{n-2}$$

Where:

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- Y_i / X_i Are the actual values.
- Y_i are the predicted value from the regression equation.
- n is the number of observations.

Let's calculate these steps:

- Compute the predicted values $\hat{Y}_i$ for each X_i.
- Compute the square difference $\left(Y_i - \hat{Y}_i\right)^2$
- Sum these squared differences.
- Divided by $n-2$ and take square root.

Using the result:

$$SE = \frac{\sum\left(Y_i - \hat{Y}_i\right)^2}{15-2}$$

Standard Error of Estimate:

The calculations result in:

- Sum of square errors: 587.11
- Standard Error of Estimate: 6.72

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